

Programming Assignment 1

Due at the beginning of your discussion session on
September 1-3, 2026

Quiz

In addition to the following topics, the quiz syllabus includes any material covered in the lectures:

- Chapter 2: “Software Construction: Building Software” in Code Complete
- Table 3.1, “Choosing Between Iterative and Sequential Approaches”, Sections 3.3 and 3.5 in Code Complete
- Section 4.3 in Code Complete

Programming

In this assignment, you will write a method that finds the longest higher suffix in two lists:

```
static <T extends Comparable<? super T>>  
List<T> longestHigherSuffix(  
    List<T> a, List<T> b, Comparator<T> cmp)
```

A *suffix* of a list is a list containing the last entry (or entries) of the list. A higher suffix is a suffix of list *a* whose elements are greater than or equal to the corresponding element in list *b*. The method is supposed to return the higher suffix of maximum length. Here are some examples:

a	b	Longest higher suffix
0, 2, 4	1, 2, 3	2, 4
1, 2	2, 1	2
2, 4	1, 3, 2, 4	2, 4
1, 2, 3, 4	1, 2, 4	2, 3, 4
2, 1	1, 2, 3	Empty list
1, 3, 2, 4	1, 2, 3, 4	4

To make the assignment more exciting:

- If your CWRU network id ends with a 0, 3, 6, or 9, then your code should use `Iterators` or `ListIterators`.
- If your CWRU network id ends with a 1, 4, or 7 then your code can use recursion but cannot use any type of loop (`while`, `do`, `for`, `for-each`, `Iterator`, `ListIterator`, `Streams`, etc.)
- In all other cases, your code should use `Streams`, but cannot use any other type of loop (`while`, `do`, `for`, `for-each`, `Iterator`, `ListIterator`, etc.) or recursion, and should avoid programming loops “into” `Stream`.

Canvas Resource

The Course Document page contains links to some of the Java features that are helpful for this assignment, such as collections and streams.

Submission

Submit an electronic copy of your program to canvas.

Grading Guidelines

The first assignment is required but not graded.